IRoC-U

ISRO Robotics Challenge - URSC

Let's build a space robot

Proposal Report

By

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System Architecture

I. Overview:

The rover system shown in a block diagram form in Fig. 1 has five key subsystems and a central **onboard computer**. A powerful **Jetson Nano** crunching sensor data, executing path planning algorithms, and coordinating actuator commands have been selected for autonomous **obstacle and crater** avoidance, **color/shape-based** sample grabbing, and precise object placement, making the rover a robust exploration tool.

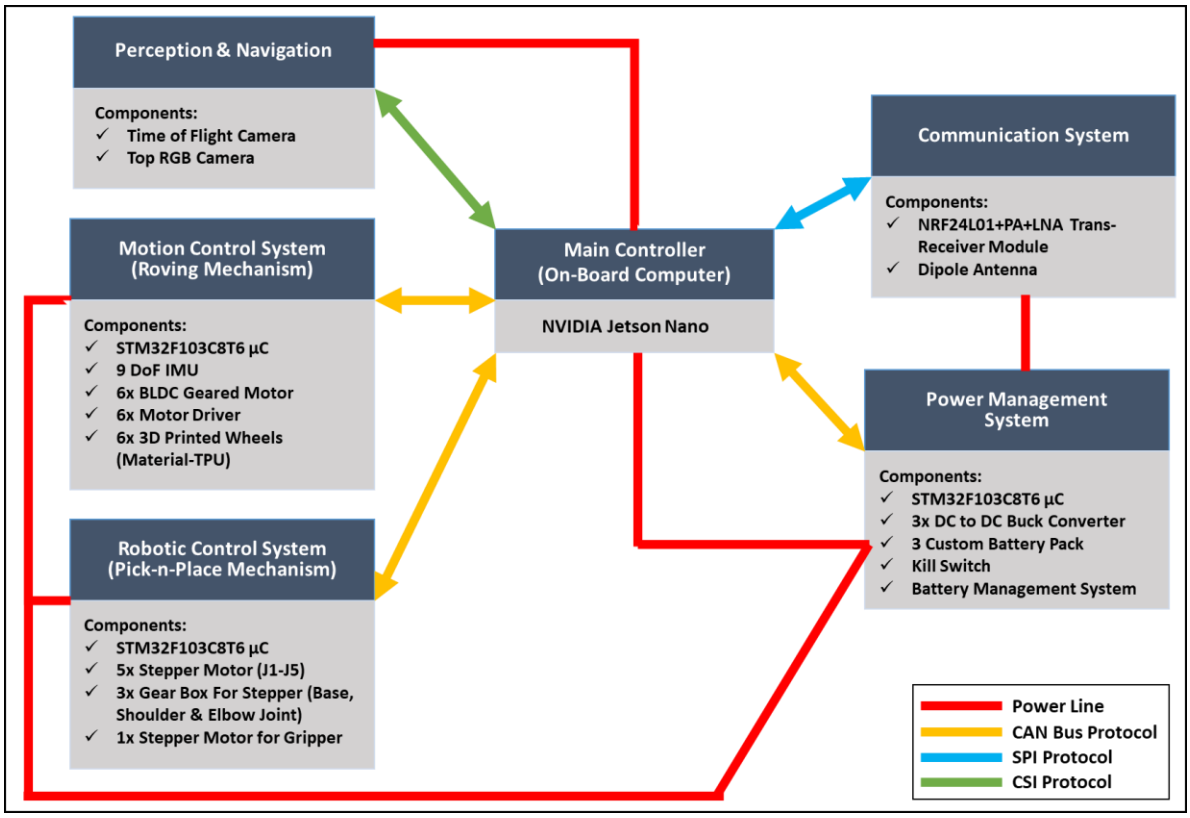


Fig. 1: Block Diagram of a Roving System with an On-Board Controller and 5 Sub-systems.

The subsystems are as follows.

II. Subsystems:

A. Perception and Navigation Subsystem:

The perception and navigation subsystem leverages a **Time of Flight camera** for front-obstacle avoidance and an **RGB camera**, along with a lightweight **CNN model** for crater detection. By fusing these two data on the **Jetson Nano**, it generates a **crater- and obstacle-free path autonomously**, which is then transmitted via **CAN** bus to the motion control system for real-time, on-path rover traversal. It also helps the **robotic arm control subsystem** with the sample object location in the real world to grab it precisely.

B. Motion Control Subsystem:

The motion control system executes path data received from the perception & navigation subsystem, using its **6-wheel Rocker Bogie** mechanism. Feedback from a **9-DOF IMU** ensures trajectory maintenance and precise positioning until reaching the designated goal. An **STM32F103C8T6 µC** is used as the brain of this motion control subsystem.

C. Robotic Arm Control Subsystem:

The robotic arm, consisting of a 5-axis serial manipulator, is controlled by an **STM32F103C8T6 µC**. It seamlessly integrates perception, navigation, and manipulation for sample collection and placement. Upon receiving target location data from the perception and navigation subsystem via **CAN bus**, the 5-axis serial manipulator, powered by **5 stepper motors**, grabs the sample object with its two-finger gripper, equipped with a servo motor and two force sensors for precise object handling. It then autonomously navigates to the designated drop location, ensuring efficient and reliable sample delivery.

D. Communication Subsystem:

The Communication Subsystem consists of one RF transceiver (**NRF24L01+PA+LNA Module - Range up to 1KM**) of **2.4 GHz** on the rover along with an **STM32F103C8T6 µC** and another at the base Station. It also uses 2 dipole antennas at both ends. This provides a dependable link for telemetry and programming the rover when necessary.

E. Power Management Subsystem:

The power system of our rover is meticulously crafted with efficiency, versatility, and robustness in mind. Using a tiered architecture, it caters to diverse power demands efficiently. **Tier 1** powers vital electronics with a dedicated **12V 12500mAh** battery, ensuring critical functions like cameras, sensors, microcontroller, and onboard computer. **Tier 2**, a robust **24V 20000mAh** battery, fuels the robotic arm's precise movements and sample manipulation. Tier 3, a formidable 24V 25000mAh battery, drives the rover's mobility with powerful gear motors. Tier 4 employs **buck converters** for seamless power distribution and regulation. An **emergency kill switch** ensures immediate circuit severance for safety. This tiered approach enhances efficiency, versatility, reliability, and safety.

Roving System

I. Justification of Selection:

A low-weight open-chassis roving vehicle shown in Figure 2 has been planned. It consists of independent drive for each of the six wheels in a rocker-bogie suspension system suitable for terrain with varying slopes along with occasional obstacles and craters. The innovation of separate drives and four encoders will ensure low wheel slip and better utilization of the battery power. On the suspension system, a robotic manipulator is planned to be mounted on a platform above the center of the front four wheels for collecting samples from the foreground. Towards the rear of the vehicle platform, the battery will be placed.

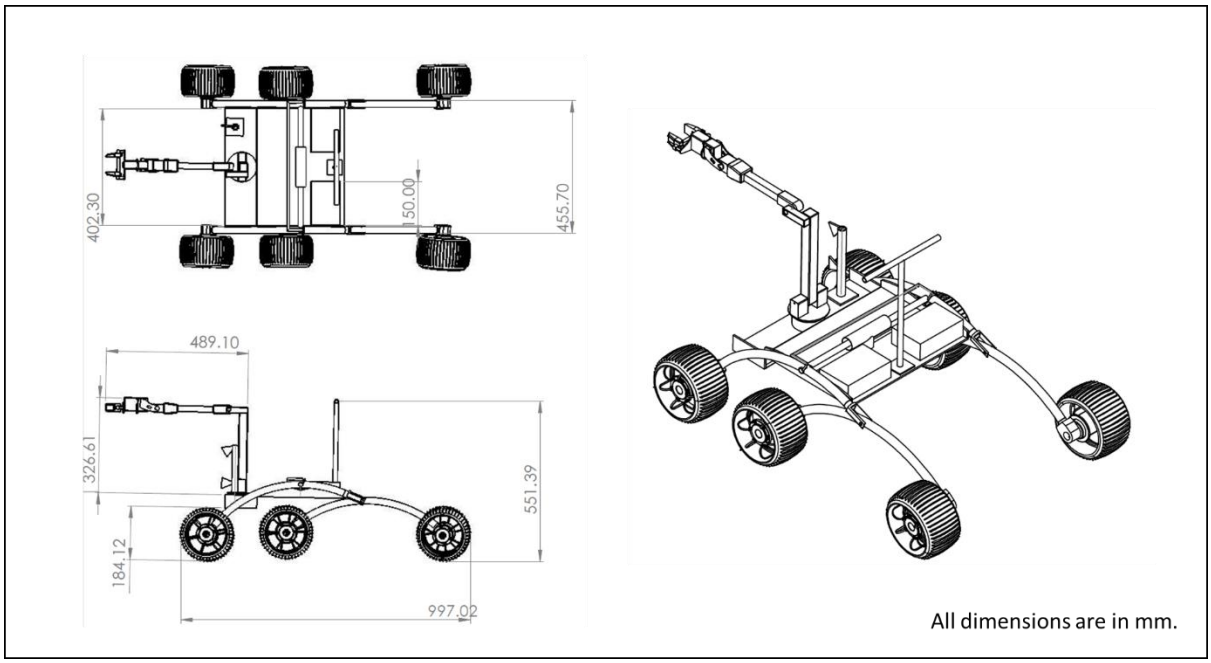


Fig. 2A: A 2D open-chassis roving vehicle with a six-wheel rocker-bogie suspension system carrying a robotic manipulator

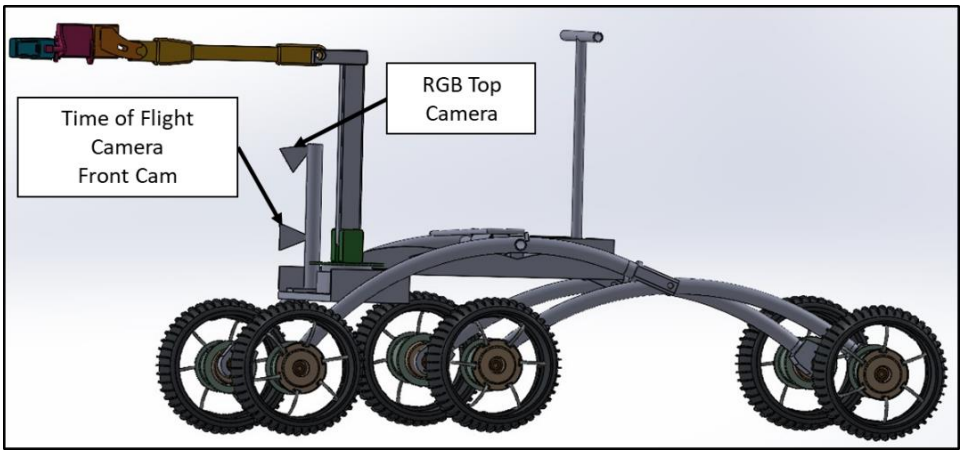


Fig. 2B: Side view of the roving vehicle

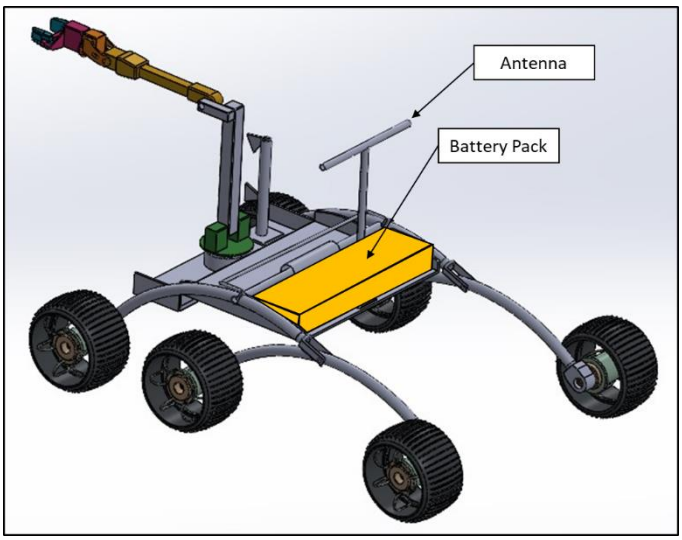


Fig. 2C: Isometric view of the roving vehicle

II. Description:

The rear pair of wheels on each side rock together about the rear hinge axis laid across the vehicle and the front wheel independently about a parallel hinge axis in front. Thus, the rear two wheels on each side establish ground contact on a plane that may or may not contain the line of contact of the front wheel. Separate control of the rotation of the wheels would ensure equal components along or parallel to this plane for the linear ground velocities due to the wheel rotation at the respective contact lines. This equality keeps the wheel slip to the minimum while negotiating a curved slope under the constraints of the constant center-to-center distance of the rear wheel pair and the constant velocity of the rigid platform.

A contribution to the ground velocity addition that due to rotational wheel drive emerges passively on the front wheel by its rocking that is different from the two wheels behind it. During negotiating a discontinuous slope change or a curved slope, the differential rocking sets the front wheel at an appropriate distance from the middle wheel with respect to that on the ground with no slope change.

An innovation of using encoders on the two hinge axes of the suspension mechanism on each side will be incorporated to estimate the angle to establish the equality of the speed component of the rear wheel with the front pair. By estimating the positions of the front two wheels on the terrain observed by the camera, the slopes at the ground at their line of contact will be determined. These angles are necessary for executing the first quality described above about the front and middle wheels.

Independent control of the wheel speeds will be used also for steering the vehicle through large obstacles on the terrain.

Pick-and-Place Robotic Mechanism

I. Justification of Selection:

A 5R robotic arm designed with five revolute or R joints, each powered by an actuating motor, is depicted in Fig. 3. The design targets a compact shape in a folding position along with optimization of sample handling efficiency and stability of the vehicle during both sample pick-and-place and slope negotiation. While picking the sample up from the ground, the arm mounted on a platform above the center of the four front wheels, as depicted in Fig. 2, would stretch ahead from the stationary rover. Of course, the battery position has been decided towards the rear side of the chassis to ensure stability at the most stretched position of the arm. A full rotation of the arm about the mounting hinge normal to the platform away from the ground would be utilized for stabilizing the rover while negotiating any sloping track. Since such a full-rotation revolute joint is also called a twisted joint, the alternate terminology for the 5R configuration contemplated here with limited-angle rotation for the inner three joints of the serial arrangement can also be named TRRRT configuration. A kinematic drawing of the manipulator is given in Fig. 4.

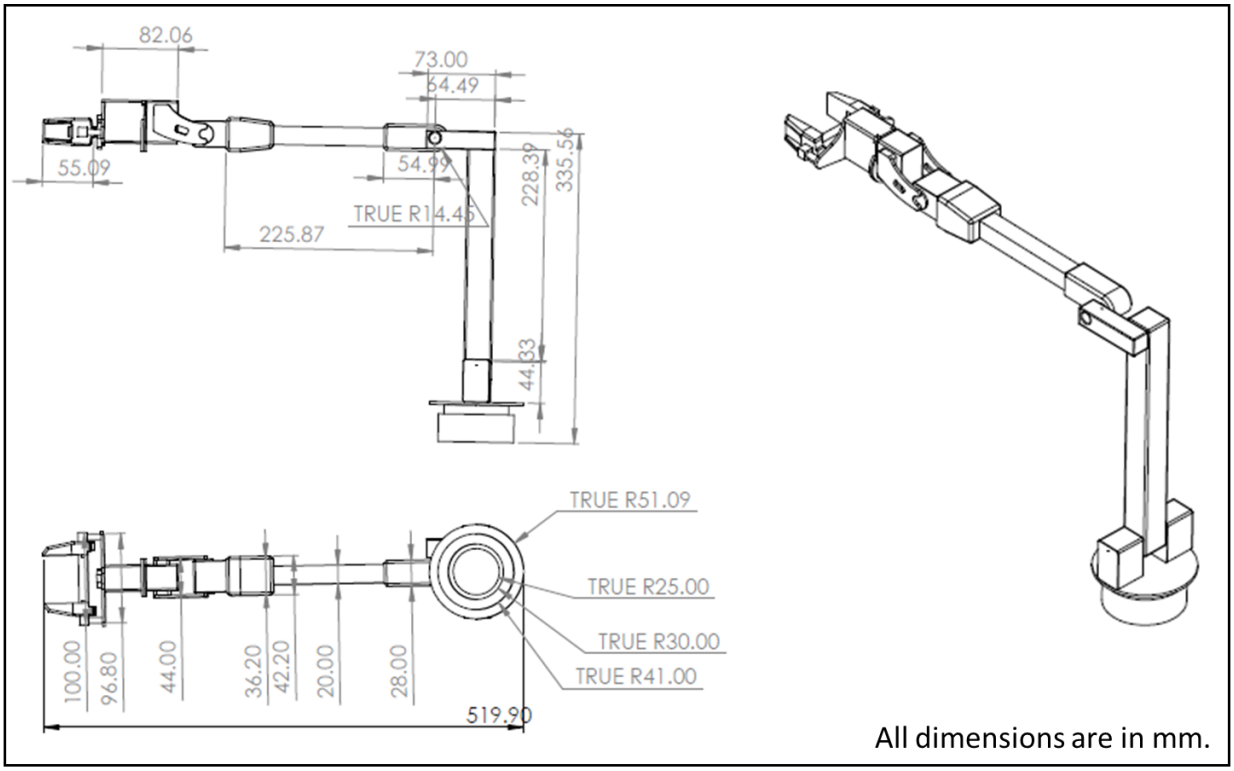


Fig 3: CAD Model of 5R, TRRRT, Robotic Arm.

II. Component Selection:

The selected actuators with encoders provide a closed-loop control with precise positioning of each joint. STM 32 microcontroller has been chosen for actuator motion control. Following position data from the Jetson Nano controller, the STM32 orchestrates the motors of the robotic arm to stretch out and grip the target object. The motors have been selected based on the torque requirement. The torque requirement of the actuators has been presented in **Tables 1** and **2**. Two force sensors have been placed at the gripper for handling the object safely. Link lengths have been optimized to ensure compliance with Arena Guidelines and efficient task completion.

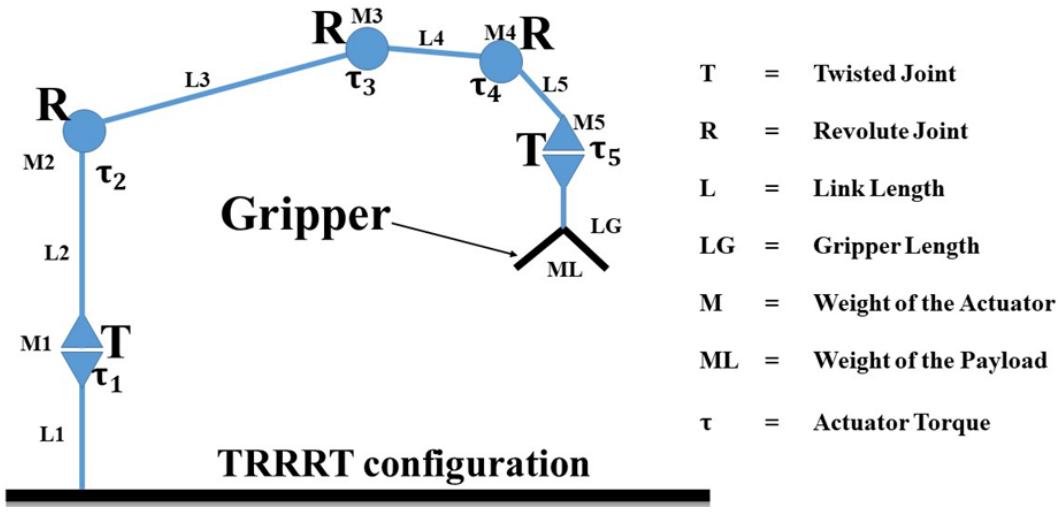


Fig 4: Kinematic Diagram of Serial TRRRT Robotic Manipulator.

Table 1: The length of the links	
Link	**Length (mm)**
L1	75.7
L2	228.39
L3	248
L4	84.05
L5	82.14
LG	62.85

Table 2: Torque Calculation		
	Joints	**Kg cm**
Actuator torque	τ_1	32.562
	τ_2	28.1629
	τ_3	23.1584
	τ_4	18.8106
	τ_5	8.5094
Gripper torque		1.5483

Emergency Response System

All the Emergency Situation & the Response Strategies are given below.

Emergency	Description	Response Strategy
Motor failure	One or more motors stop working, causing the rover to stall or move erratically.	Verify power, connections, and motor status. Switch to functioning motors. Safely stop the rover with the emergency brake. Send a distress signal for assistance.
Sensor malfunction	Sensors providing crucial data (e.g., camera, IMU) malfunction	Utilize redundant sensors, calibrate for accuracy, modify navigation algorithms based on available data, and alert the base station for assistance.
Communication loss	Connection with the base station is lost, hindering command reception and data transmission.	Retry communication via alternative channels, store critical data onboard, activate autonomous mode, and search for stronger signals in suitable areas.
Battery depletion	The rover's battery runs low, risking shutdown or loss of essential functions.	Optimize energy use by shutting down non-essential systems, find a charging station, enter low-power mode, and signal the base station for assistance.
Software failure	Software bugs cause system instability.	Restart the system to dissolve glitches, run diagnostics to troubleshoot, load backup for stability, and seek remote assistance for debugging and updates.
Abnormal Behavior of the Rover	If the rover is in such a situation where human intervention is required (Wheels begin to spin furiously, drawing excessive current and triggering motor overheat warnings)	Stop all the function of the rover by pressing the emergency kill switch, repair the problem and restart the rover by releasing the kill switch.

Hardware Identification

All the hardwares listed below are commercial off-the-shelf items.

I. List of Hardware Required (Subsystem wise):

Perception & Navigation Subsystem			
Hardware	**Category**	**Quantity**	**Justification**
Camera	Raspberry Pi Camera V2	1	High-resolution Pi Cam V2's 8MP sensor delivers detailed images crucial for accurate crater identification.
	Arducam Time of Flight Camera	1	This ToF camera is for obstacle detection and avoidance by using the depth of the image.
Motion Control Subsystem			
9-DOF Inertial Measurement Unit	MPU9250	1	The MPU9250's compact size integrates a gyroscope, accelerometer, and magnetometer, providing precise motion tracking crucial for accurate rover movement and orientation estimation.
BLDC Geared Motor	High Torque DC Gear Motor 24V,58KG-Cm	6	High torque DC gear motors offer a reliable and powerful motion for the Rocker Bogie mechanism
Motor Driver	20 Amp 6V-30V BLDC Motor Driver (2 Channels)	6	The 20A 6V-30V DC motor driver's high current handling capacity and wide voltage range ensure efficient and robust power delivery.
Microcontroller	STM32F103C8T6	1	STM32F103C8T6's balance of processing power for motor control peripherals & cost-effectiveness makes it ideal for the Brain of the motion control system.
CAN Bus Controller	TJA1050 CAN Bus Controller	1	High-speed, reliable CAN bus interface & robust isolation for reliable communication between microcontroller and Jetson Nano.
Wheel	3D Printed with TPU95A Material	6	TPU 95A's high resilience, abrasion resistance, and shock absorption make it ideal for 3D-printed rover wheels, handling diverse terrain effectively.

Robotic Arm Control Subsystem			
Hardware	**Category**	**Quantity**	**Justification**
Stepper Motor	Nema 23 with 5:1 Gearbox x 3 Nema 17 x 2	5	Combining precise control, high torque, and NEMA 17 & 23 stepper motors offers the perfect for robotic arm movements.
Stepper Driver	DM542 Digital Stepper Motor Driver	5	DM542's advanced DSP control and bipolar constant-current chopping deliver smooth, precise arm movements with low noise and heat.
Two-Finger Gripper	DS3240 Pro Servo Motor	1	Enables grasping and holding objects with varying shapes and sizes.
Force Sensor	Force-sensitive resistor	2	Provides feedback on applied force to ensure safe object handling.
Microcontroller	STM32F103C8T6	1	STM32F103C8T6's balance of processing power for motor control peripherals & cost-effectiveness makes it ideal for the Brain of the robotic arm control subsystem.
CAN Bus Controller	TJA1050 CAN Bus Controller	1	High-speed, reliable CAN bus interface & robust isolation for reliable communication between microcontroller and Jetson Nano.
Communication Subsystem			
2.4 GHz Radio Frequency Transceiver	NRF24L01+PA+ LNA	2	Combining long-range thanks to the Power Amplifier, low power consumption, robust performance with Low Noise Amplifier
Antenna	Di-Pole Antenna	2	High-gain Omnidirectional antennas for long-range communication with the base station.
Microcontroller	STM32F103C8T6	1	STM32F103C8T6's balance of processing SPI peripherals & cost-effectiveness makes it ideal for the Brain of the Communication System.
Computation (Main Controller)			
Single Board Computer	Jetson Nano	1	Jetson Nano's powerful AI processing tackles complex perception, planning, and control tasks crucial for intelligent rover autonomy.
CAN Transceiver	MC2515 CAN bus Module	1	This CAN in bus transceiver is used for communicating with all other subsystems

Power Management System			
Hardware	**Category**	**Quantity**	**Justification**
Custom Li-Po battery pack Battery	• 12V,12500 mAh • 24V,25000 mAh • 24V,20000 mAh	3	High capacity and long-lasting battery for extended operation. 3 Battery Packs for 3 vital sections namely Electronics Section, Robotic Arm Section, Rocker Bogies Section
DC-DC Buck Converter	1x 12V to 3.3V, 5V 2x 24V to 3.3V, 5V	3	Efficiently steps down voltage, safeguarding sensitive circuits from overvoltage damage.
Battery Management System	12V x 1 24V x 2	3	Prevents battery fires, extends life, optimizes performance
Kill Switch	24V, up to 300 A	1	Emergency stop for safety, preventing damage or injury.
Mechanical Frame			
Rocker Bogie Links	Carbon Fiber Square Hollow Tube	2	High Weight to Strength Ratio
Robotic Arm Links	Carbon Fiber Round Hollow Tube	2	High Weight to Strength Ratio
Chassis	Aluminum Alloy Tube	4	High Rigidity to Weight Ratio
Enclosure for Electronics	IP65 Enclosure	2	Water & Air Proof

Software Identification

To accomplish the mission objectives, software subsystem design and implementation are crucial. A detailed list of software subsystems needed for the required mechanisms is provided below.

I. Perception & Navigation Software Subsystem:

A. Algorithms:

1) Obstacle Detection Algorithm:

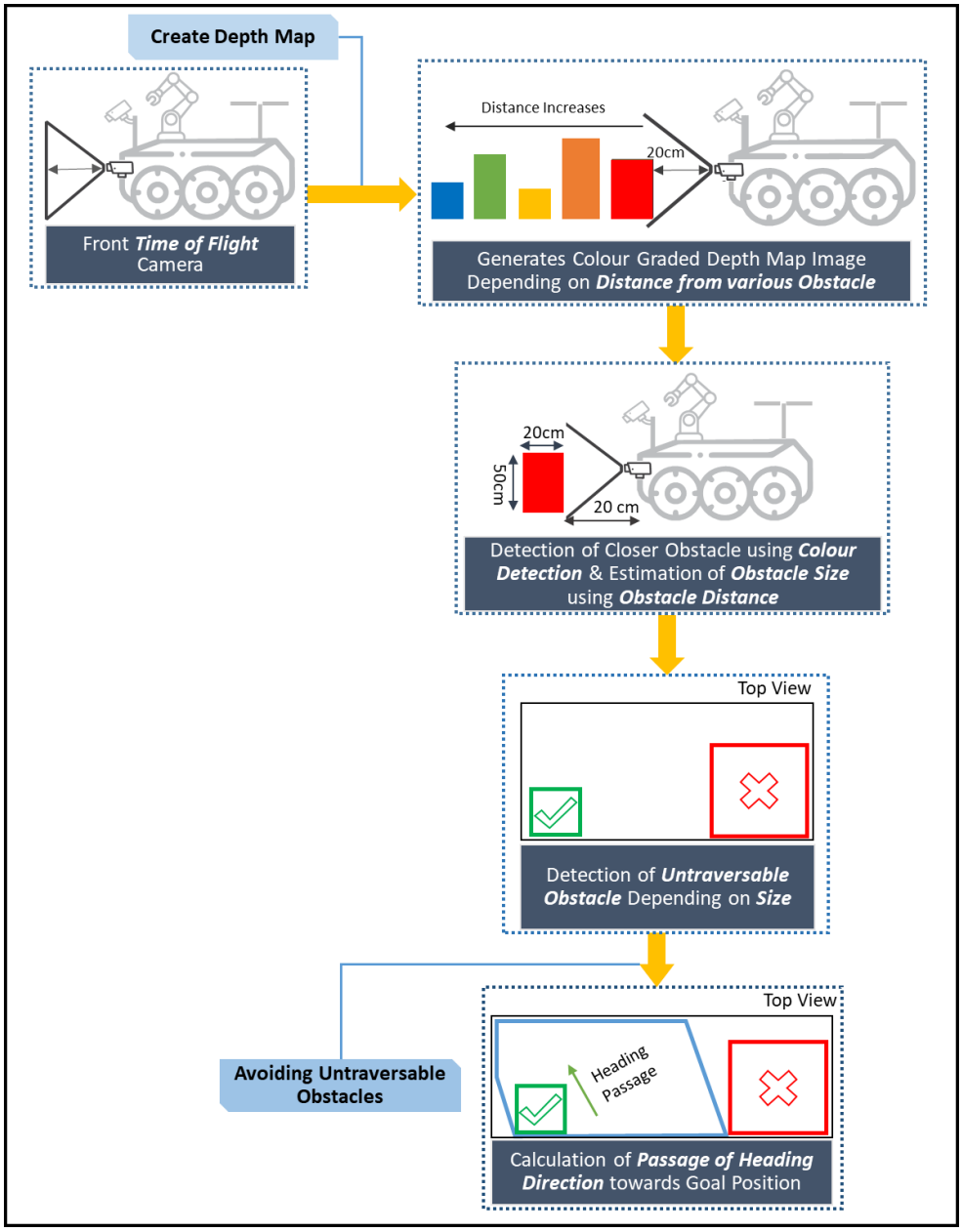


Fig5: Obstacle Detection Algorithm Flow Chart

2) Crater Detection Algorithm:

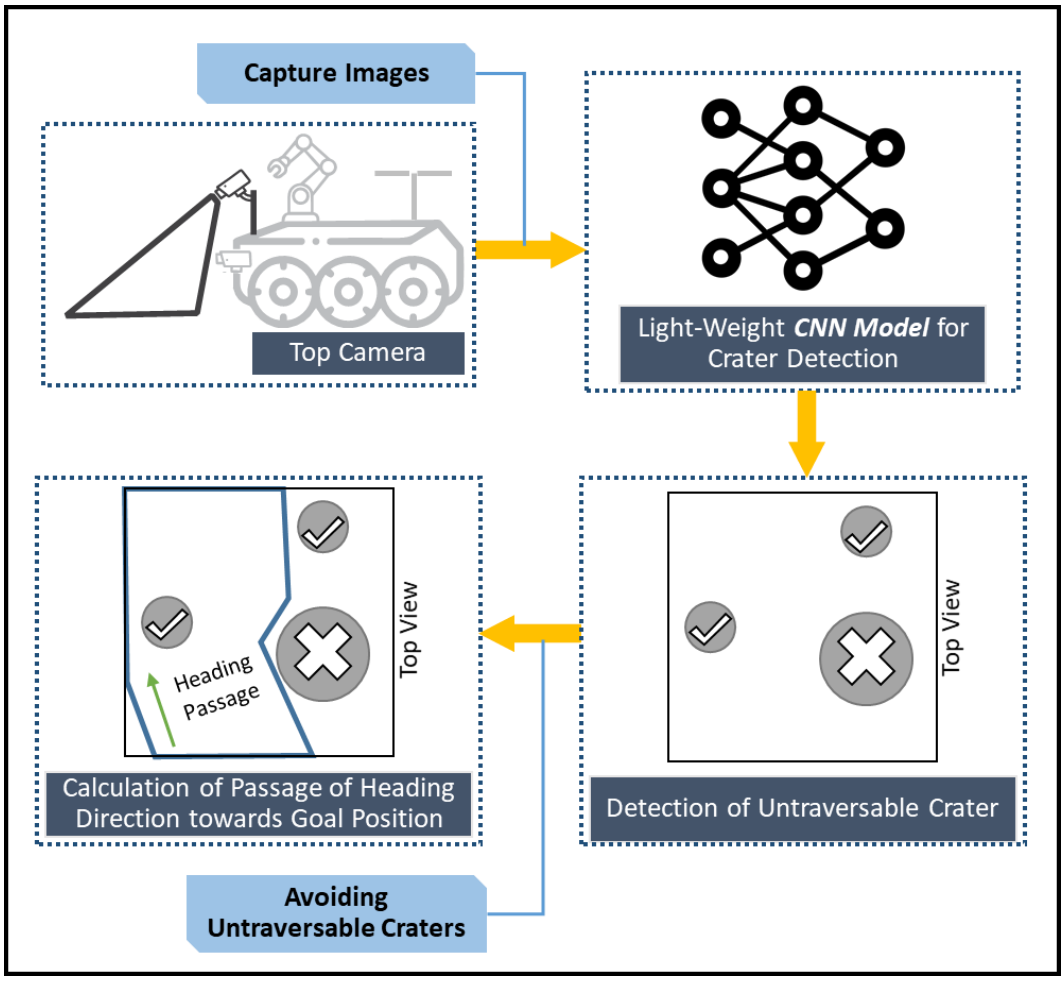


Fig6: Crater Detection Algorithm Flow Chart

In **Figure 5**, the **time-of-flight camera** gauges distances, creating a **color-graded depth map image**. By applying color detection to this image, the rover identifies the nearest **obstacle**, allowing **avoidance**. Additionally, it determines **obstacle size** based on distance, aiding in decision-making regarding which obstacles to avoid. This process ultimately guides the rover in finding the **heading passage** through the environment.

In **Figure 6**, the rover's top camera employs a **lightweight Convolutional Neural Network (CNN)** model, executed on the **Jetson Nano** Single Board Computer. This sophisticated setup allows the rover to detect craters accurately. The CNN model analyzes visual data, providing essential information to determine a clear heading passage for the rover. This integration of advanced image processing with the computational power of the Jetson Nano enhances the rover's ability to navigate autonomously.

3) Path Planning Algorithm:

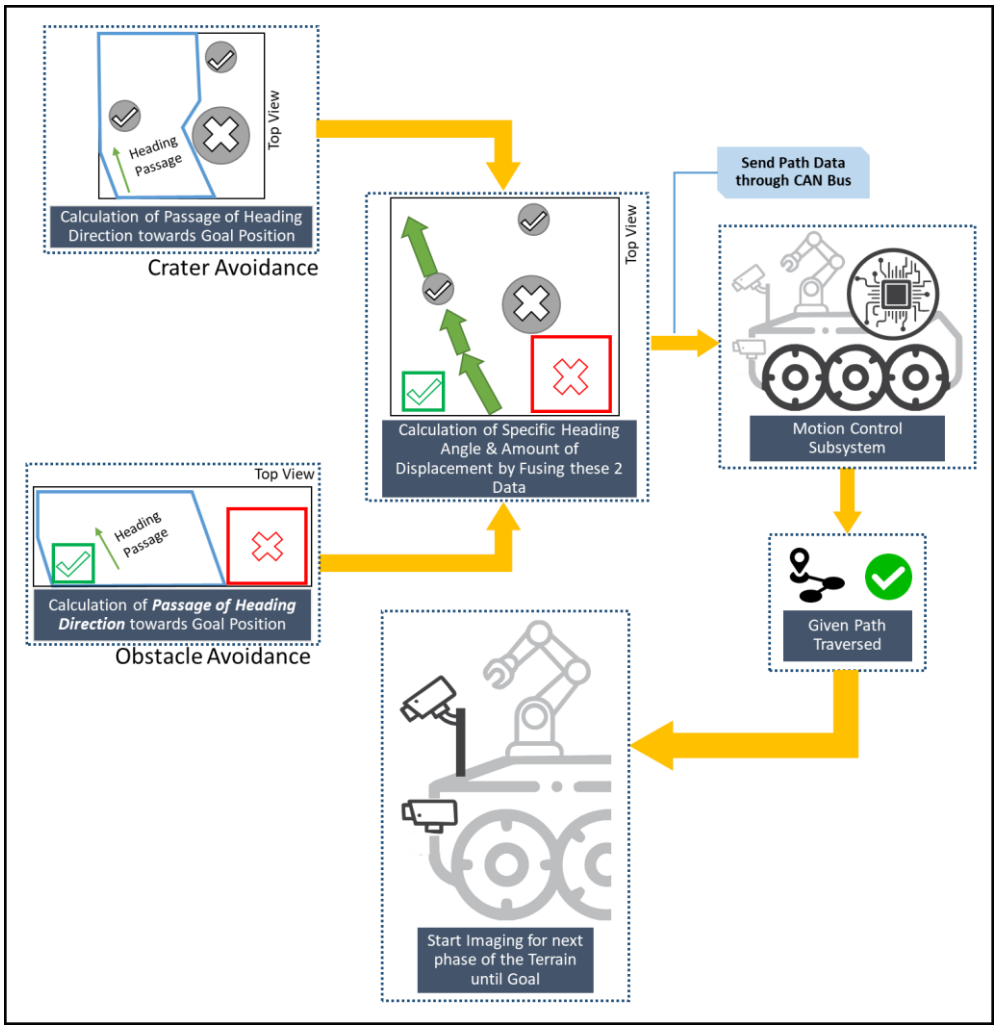


Fig7: Path Planning Algorithm Flow Chart

In **Figure 7**, the fusion process involves integrating the **obstacle-free heading passage** with the **crater-free heading passage**, resulting in the determination of a **specific angle** towards which the rover should move. This calculated path data is then transmitted to the motion control subsystem via **CAN bus protocol**, guiding the rover along a trajectory that avoids both obstacles and craters.

B. Development Environment:

- **Hardware:** Jetson Nano, One Front Time Of Flight camera and One Top RGB camera.
- **Software:** Python-OpenCV for image processing, TensorFlow Lite for CNN model.

II. Motion Control Software Subsystem:

A. Algorithm:

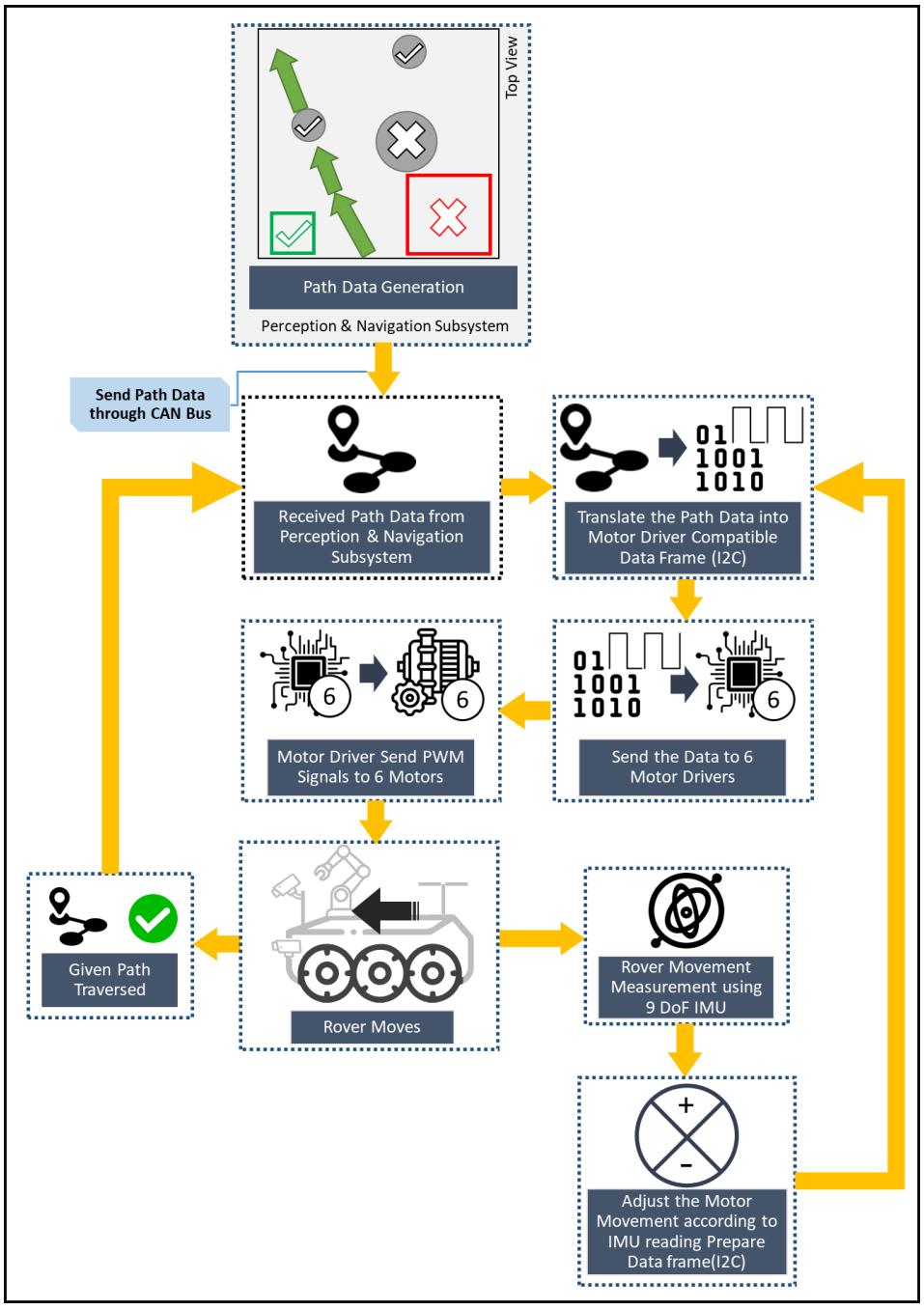


Fig8: Motion Control Software Flow Chart

In **Figure 8**, upon receiving path data from the perception and navigation subsystem, the motion control system initiates the execution using the 6-wheel Rocker Bogie mechanism. It translates the path data into a motor driver-compatible data frame using I2C communication, sending the data to the motor drivers. The motor drivers transmit PWM signals to the BLDC

motors for rover movement. The 9 degrees of freedom inertial measurement unit activates during rover motion, providing feedback signals to the controller, ensuring the rover stays on course by adjusting acceleration, position, and orientation.

B. Development Environment:

- **Hardware:** STM32 microcontroller, 6 BLDC geared motor and their drivers, 9 DOF IMU, TJA1050 CAN Controller.
- **Software:** STM32 cube IDE for the STM32micro controller, libraries for motor driver communication, sensor fusion libraries for IMU data integration, and CAN communication libraries.

III. Robotic Control Software Subsystem:

A. Algorithms:

1) Pick Operation:

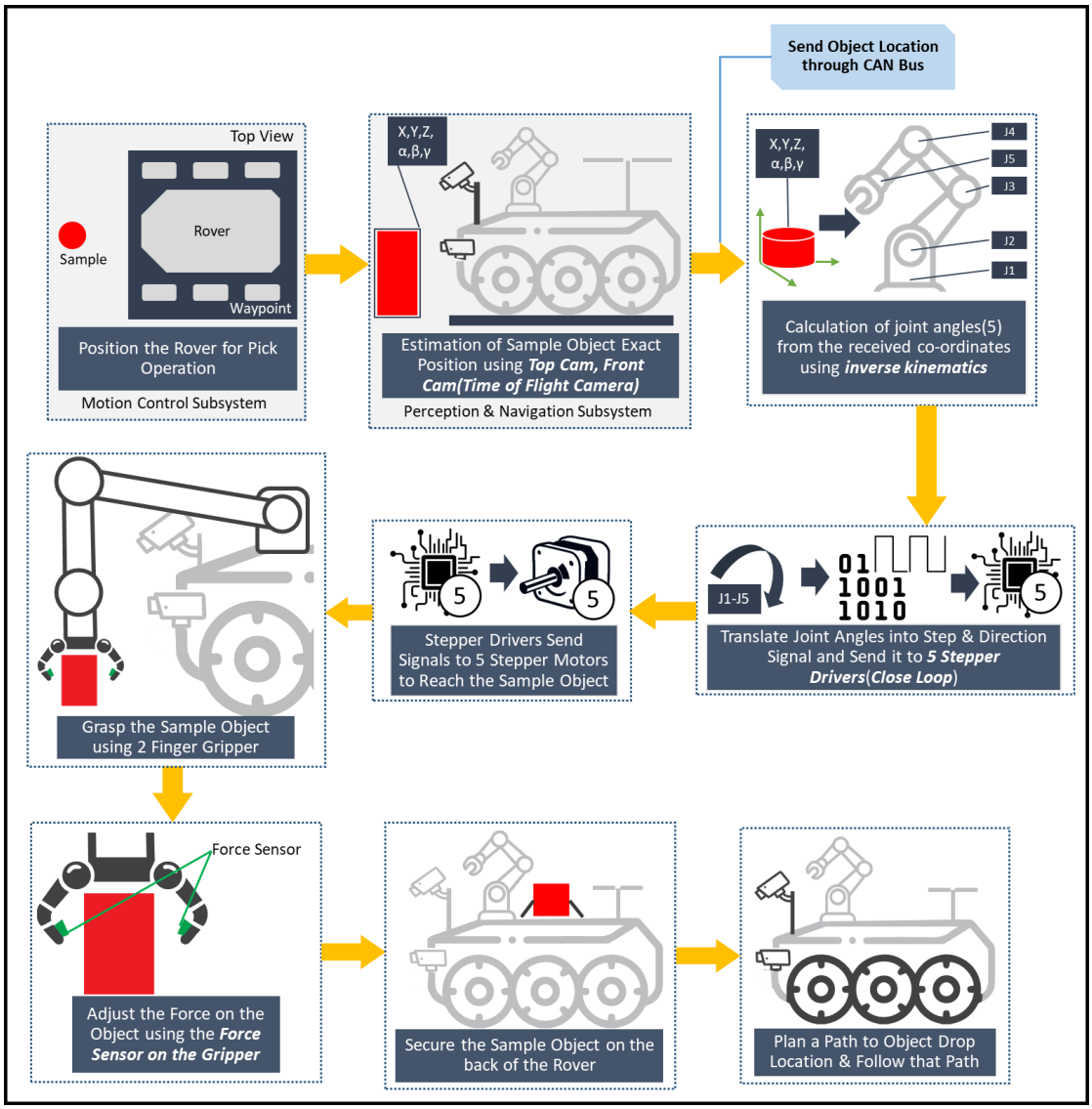


Fig9: Robotic Arm Pick Operation Flow Chart

2) Place Operation:

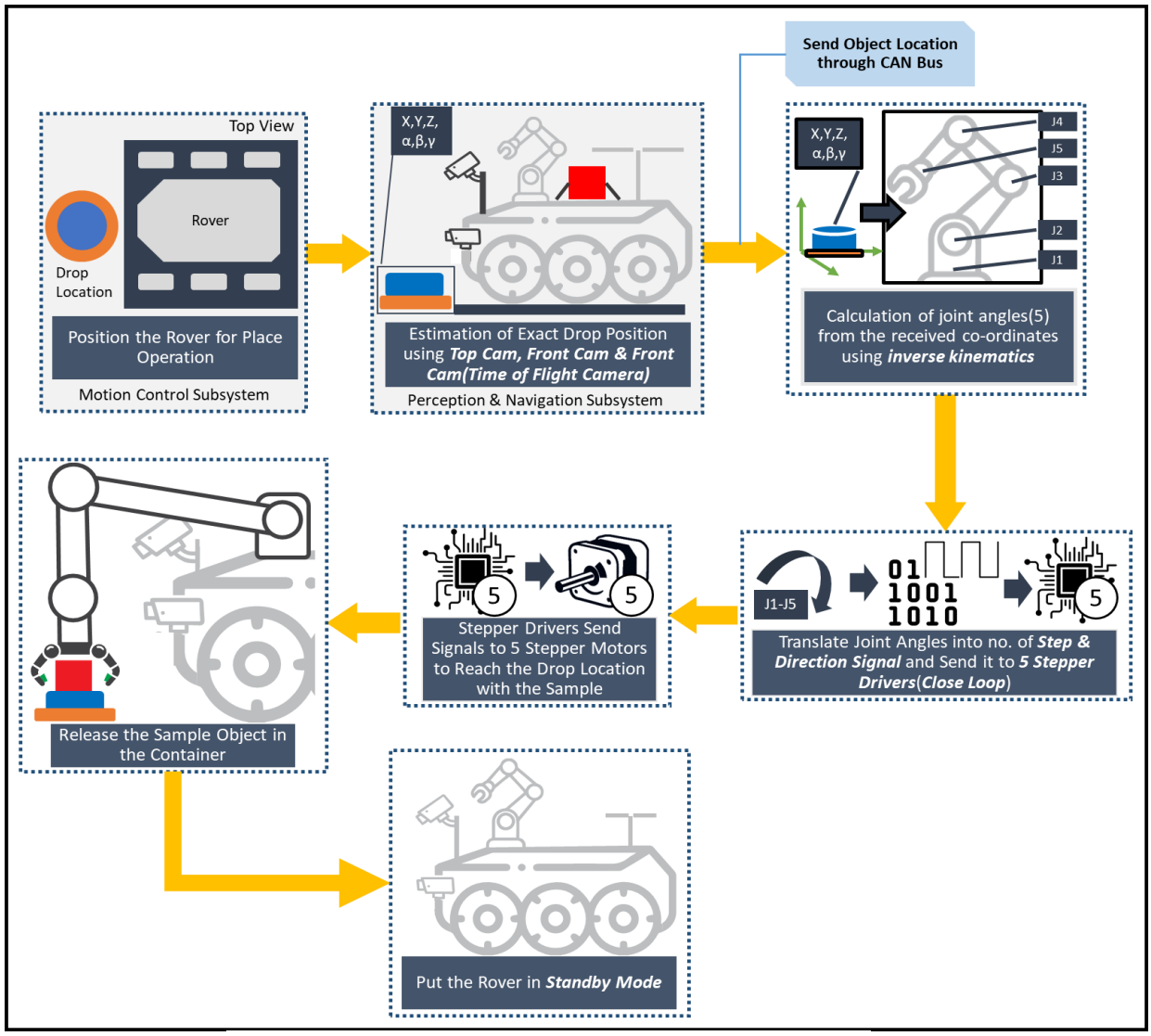


Fig10: Robotic Arm Pick Operation Flow Chart

In Fig.9 & Fig 10, after reaching a waypoint, the rover carefully positions itself using the **motion control subsystem** to grab a **sample object**. The **perception and navigation subsystem** precisely locates the sample using **time-of-flight and RGB cameras**, transmitting the **XYZ coordinates** via **CAN bus** to the **robotic arm control software**. The robotic arm, directed by **stepper motors**, moves to the **sample location**, grabs it with adequate force, using **force sensor feedback**, secures it on the rover's back, and autonomously drives. The rover **visually identifies the drop location**, then carefully **places the sample object** at the designated spot.

B. Development Environment:

- **Hardware**: STM32 microcontroller, 5 stepper motors along with drivers, TJA1050 CAN Controller.
- **Software**: STM32 cube IDE for the STM32 microcontroller, libraries for motor driver communication, libraries for IMU data integration, and CAN communication libraries.

IV. Communication Software Subsystem:

A. Algorithms:

1) Base Station:

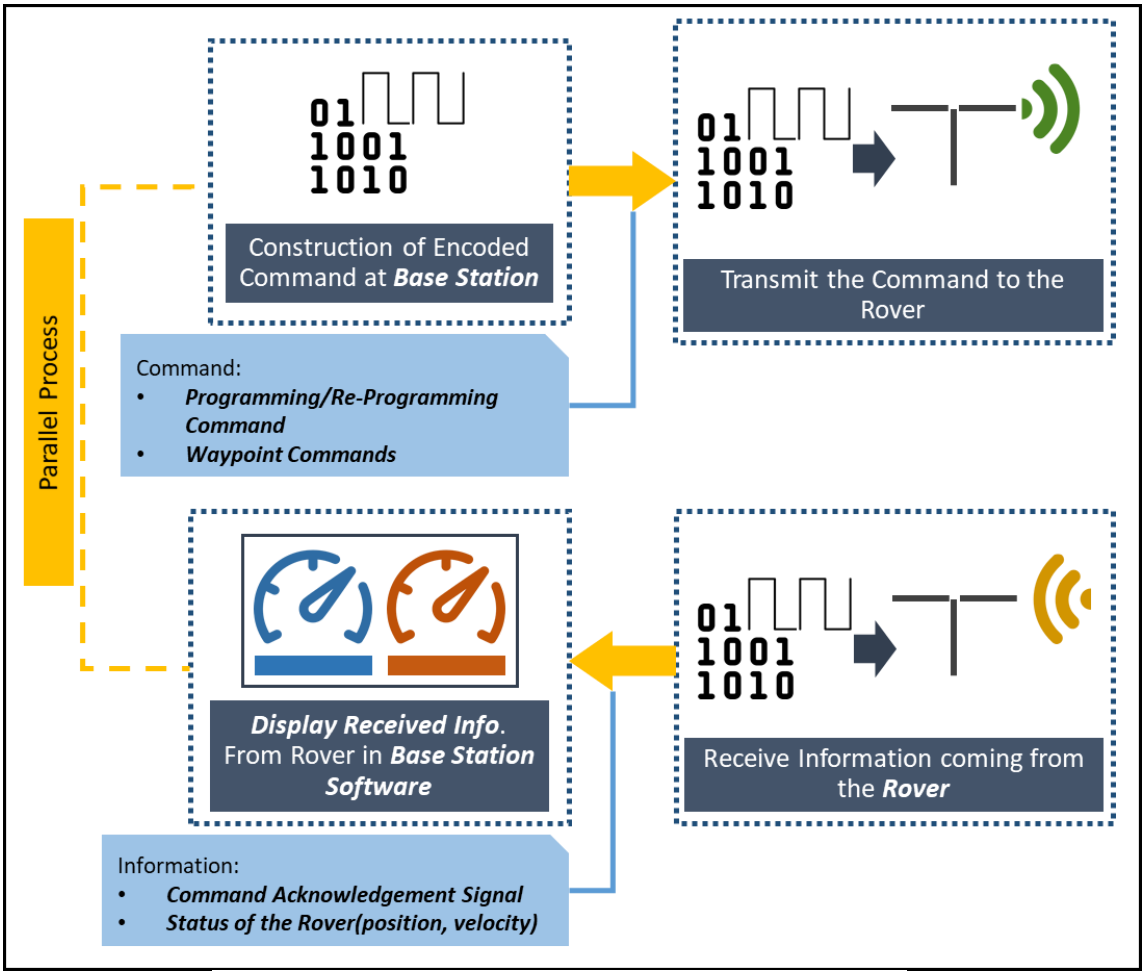


Fig11: Base Station Communication Flow Chart

In **Figure 11**, the command is formulated at the base station using dedicated software and transmitted via the **RF transceiver**, equipped with a **dipole antenna**. Simultaneously, the base station receives real-time rover status information, displayed comprehensively on the station's interface. This fundamental algorithm outlines the seamless interaction between the base station software, the RF transceiver, and the rover, facilitating bidirectional communication, command transmission, and status feedback for effective control and monitoring within the rover system.

2) Rover:

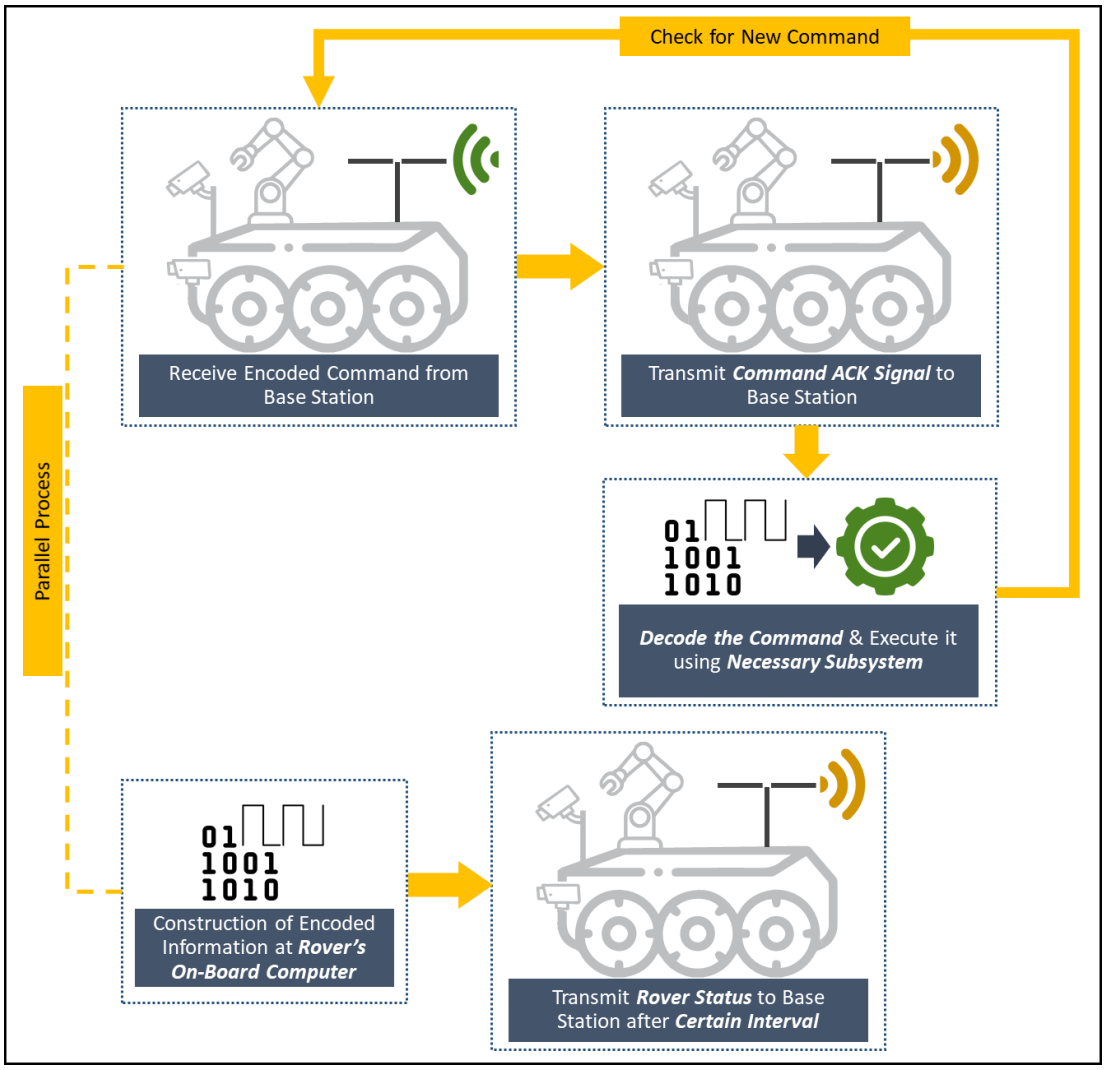


Fig12: Rover Communication Flow Chart

In **Figure 12**, the rover communication subsystem actively monitors the base station for **new commands**. Upon receiving a command, the rover promptly **acknowledges the signal** to the base station and **executes the command**. Simultaneously, at regular intervals, the communication subsystem autonomously dispatches the current **rover status to the base station**.

B. Development Environment:

- **Hardware**: 2 NRF24L01+PA+LNA Modules, Dipole Antenna, 1 STM32 microcontrollers (for rover).
- **Software**: STM32 cube IDE for the STM32 microcontrollers, libraries for RF transceiver.

Hardware and Software Realization Plan

I. Hardware Realization Plan:

Hardware Details	Procurement Source	Specifications/ Realization Plan	Quantity	Estimated Cost (INR)
Raspberry Pi Camera V2	Market	8-megapixel Sony IMX219 sensor, 1080p30 video recording, 15cm ribbon cable, Resolution: 3280 x 2464 pixels	1	₹ 2,000.00
Arducam Time of Flight Camera	Market	Depth range up to 4m, ideal for obstacle avoidance and environment mapping.2MP image sensor for simultaneous color and depth data.	1	₹ 5,700.00
MPU9250 -Inertial Measurement Unit	Market	Gyroscope, accelerometer, and magnetometer, 16-bit resolution, I2C interface	1	₹ 1,200.00
High Torque BLDC Gear Motor 24V,58KG-CM	Market	24V power supply, 58kg load capacity,,20mm diameter shaft	6	₹ 70,000.00
BLD300 Brushless DC Motor Driver	Market	20A continuous current rating, 6V-30V input voltage, 4-pin interface	6	₹ 23,000.00
STM32F103C8T6	Market	ARM Cortex-M3 core, 72 MHz clock speed, 128KB RAM, 512KB Flash memory, 64-pin LQFP package	3	₹ 600.00
Wheel	3D Printed with TPU95A Material	CAD MODELING	6	₹ 20,000.00
Nema23 with 5:1 Gearbox	Market	3.4Nm holding torque, 56mm flange diameter, 8-wire lead	3	₹ 15,000.00
Nema 17	Market	1.7Nm holding torque, 35mm flange diameter, 4-wire lead	2	₹ 2,400.00
DM542 Digital Stepper Motor Driver	Market	Power supply: DC 24-60 (V), Current maximum: 4.5A, Output phase current:1 .5A-4.5A	5	₹ 6,000.00
DS3240 Pro	Market	40kg-cm force range, 5V power supply, analog output	1	₹ 1,000.00

Hardware Details	Procurement Source	Specifications/ Realization Plan	Quantity	Estimated Cost (INR)
Force sensitive resistor	Market	Actuation Force: 0.1 Newtons, Force Sensitivity Range: 0.1 - 10.02 Newtons, Non-Actuated Resistance: 10M W	2	₹ 600.00
NRF24L01+PA+LNA	Market	2.4 GHz frequency band, 100mW transmit power,100dB receive sensitivity, SPI interface	2	₹ 350.00
TJA1050 CAN Bus Controller	Market	High-speed (1 Mbps) CAN transceiver: supports robust automotive communication; low EMI/EMC for reliable data transmission.	3	₹ 300.00
Jetson Nano	Market	Quad-core ARM CPU, 128-core Nvidia GPU.4GB RAM, 32GB eMMC	1	₹ 15,000.00
Dipole Antenna	Market	2.4 GHz frequency band,	2	₹ 200.00
Battery	Market	• 12V,12500 mAh • 24V,25000 mAh • 24V,20000 mAh	3	₹ 74,000
DC-DC Buck Converter	Market	• 1x 12V to 3.3v, 5V • 2x 24V to 12V 5V	3	₹ 600
Battery Management System		• 12V x 1 • 24V x 2	3	₹ 3,000
Carbon Fibre Square Hollow Tube	Market	OD 30 MM / 2MM THICKNESS	2	₹ 6,000.00
Carbon Fibre Round Hollow Tube	Market	OD 30 MM / 2MM THICKNESS	2	₹ 6,000.00
Aluminum hollow pipe	Market	T-SLOT(20x20mm)	4	₹ 1,200.00
IP65 Enclosure	Market	80X74X52 MM	2	₹ 5,000.00
Total Cost				**₹ 2,59,150.00**

II. Software Realization Plan:

Perception and Navigation software subsystem				
Purpose	**Software Stack**	**Programming Language**	**Development Tools**	**Notes**
Obstacle & Crater Detection Algorithm, Fusion algorithm for path planning	OpenCV, TensorFlow Lite	Python	Visual Studio Code	Obstacle & Crater detection data must be fused carefully
Motion Control software subsystem				
Implement mobility control algorithms	PID control algorithms	C/C++	STM32CubeIDE	PID parameters needs to be tuned
Robotic Arm Control software subsystem				
Purpose	**Software Stack**	**Programming Language**	**Development Tools**	**Notes**
Control individual joints of the robotic arm for precise manipulation, with force feedback	Inverse kinematics algorithms,	C/C++	STM32CubeIDE	Arm calibration and sensor calibration are crucial
Communication Subsystem				
Establish reliable communication with the base station	Encoding & Decoding algorithm while transmitting & receiving	C/C++	STM32CubeIDE	Security and data integrity need to be addressed.
On-board Computer				
Provide a platform for image processing, control algorithms, and communication protocols.	Linux4Tegra	Python/C++	Development tools for Linux environment	Optimization and resource management are crucial for efficient operation.
Base Station User Interface				
Provide a user friendly interface for monitoring the rover's status, and for sending commands, and visualizing data	Web development frameworks (e.g., Flask), Graphical user interface libraries(e.g.Qt)	Python JavaScript	Visual Studio Code	Usability and user experience should be prioritized.

Test Plan

I. Overview:

The tests will include simulations of potential failures and emergencies, aiming to uncover any weaknesses in the rover's design and functionality. By subjecting the rover to rigorous testing conditions, we can ensure its ability to navigate and operate efficiently in a variety of challenging terrains.

II. Test Philosophy:

- **Fault Injection:** Simulate hardware/software failures.
- **Environment Emulation:** Replicate mission conditions.
- **Comprehensive Coverage:** Address all emergency scenarios.

III. Subsystem-wise Test:

Identification of the Test	Plan for the Test
Perception & Navigation Subsystem	
1. Camera functionality 2. CNN model accuracy 3. Obstacle detection and path planning 4. CAN bus communication.	1. Verify camera image quality, resolution, and frame rate. 2. Evaluate model's ability to correctly detect obstacles and craters. 3. Verify obstacle-free path generation and transmission. 4. Confirm data transfer integrity and timing over CAN bus.
Motion Control Subsystem	
1. Motor control and movement 2. Trajectory accuracy 3. IMU feedback 4. CAN bus communication	1. Test individual motor control, manoeuvres, and speed control. 2. Assess ability to follow planned paths precisely. 3. Verify IMU data usage for positioning and trajectory correction. 4. Confirm path data reception and execution commands.
Robotic Arm Control Subsystem	
1. Arm movement and dexterity 2. Gripper functionality 3. Force sensor accuracy 4. CAN bus communication	1. Test full range of motion, smoothness, and precision in reaching targets. 2. Verify gripper's ability to grasp and release objects. 3. Check force sensor readings for object detection and grip force control. 4. Confirm target location data reception.
Communication Subsystem	
1. Signal strength and range 2. Data transmission reliability 3. Telemetry and programming capabilities	1. Test signal strength and range in various environments. 2. Verify successful data exchange with base station. 3. Demonstrate ability to receive programming updates and send telemetry data.

Identification of the Test	Plan for the Test
Power Management Subsystem	
1. Battery voltage and current levels 2. Solar panel charging efficiency 3. Overcharge and undercharge protection 4. Fault detection and isolation	1. Monitor battery voltage and current draw under various loads. 2. Measure charging rate under different sunlight conditions. 3. Simulate overcharge and undercharge scenarios. 4. Trigger controlled system faults and assess detection/isolation.

IV. Subsystem Integration Test:

Test Identification	Test Plans
Perception and Navigation to Motion Control Integration	1. Verify seamless path data transfer from Perception and Navigation to Motion Control. 2. Assess Motion Control's ability to execute received paths accurately. 3. Evaluate trajectory correction based on IMU feedback.
Perception and Navigation to Robotic Arm Integration	1. Confirm target location data transfer from Perception and Navigation to Robotic Arm Control. 2. Validate Robotic Arm's ability to reach target locations precisely. 3. Assess object localization accuracy using Perception and Navigation data.
Motion Control to Robotic Arm Integration	1. Validate arm stability during rover movement. 2. Assess arm's ability to compensate for rover motion during sample collection. 3. Evaluate joint coordination for smooth, collision-free operation.
Communication Subsystem Integration	1. Verify reliable data exchange between all subsystems and base station. 2. Test remote command execution and telemetry data transmission. 3. Assess programming update capabilities via Communication Subsystem.
Power Management Integration	1. Verify power distribution to all subsystems under various loads. 2. Assess system behaviour under low-power conditions. 3. Test power management strategies for extended mission duration.
Integrated System Testing	1. Simulate complete mission scenarios, including obstacle avoidance, sample collection, and delivery. 2. Evaluate overall system performance and robustness under varying conditions. 3. Identify and address integration issues for seamless operation.

System Specifications

Specification	Value	Description
Dimensions	0.997 m × 0.651 m × 0.551m	Dimension with robotic manipulator & antenna
Mass	45 kg	
Power	1250W	From Li-Po Battery
Range	900 meters	The distance it can cover in 1 hour
Speed	40 cm/s	
Battery Lifetime	1 Hour	
Payload	• 1 RGB Camera • 1 ToF Camera • 1 Five DoF Serial Manipulator • 1 9 DoF IMU	For Autonomous Navigation & Path Planning, Sample Object Detection
All-terrain capable	Yes	Rocker Bogie Mechanism for Roving
Autonomous Navigation	Yes	Can navigate using sensors and algorithms
Sample Collection Capability	Yes	5 DoF Serial Manipulator
Robotic Manipulator Reach	650 mm	Can reach up to 300mm in -Z axis
Communication Range	1 KM	RF Link(NRF24L01+PA+LNA Module)

Project Management

I. Team Members:

Team Member	Name	Department
1	Animesh Majumder	Metallurgical And Material Engineering
2	Asfak Ali	Electronics And Telecommunication Engineering
3	Diptanshu Chakrabartty	Civil Engineering
4	Hrishitava Sarkar	Mechanical Engineering
5	Mainak Dey	Mechanical Engineering
6	Priyabrata Basu	Electronics And Telecommunication Engineering
7	Samim Akhtar (Lead)	Electronics And Telecommunication Engineering
8	Sanjay Thakur	Electrical Engineering
9	Saptarshi Maity	Metallurgical And Material Engineering
10	Sayan Kumar Sah	Metallurgical And Material Engineering

II. Project Management Table:

Values in the 2nd column and 3rd column represent the team member number from the above Table.

Task Name	Primary Task	Secondary Task	Start Date	End Date	Duration (Days)	Estimated Cost
Requirements Gathering	7	6	01-Feb-2024	15-Feb-2024	15	Nil
Final Design Review Of The CAD Model	4,5	1,3,9	01-Feb-2024	24-Feb-2024	24	Nil
Final Design Review Of The Electronics & Software System	2,6	7,10	01-Feb-2024	24-Feb-2024	24	Nil
Preparation & Submission of ISRO Rover Challenge Design Report	7	6	01-Feb-2024	29-Feb-2024	29	Nil

Task Name	Primary Task	Secondary Task	Start Date	End Date	Duration (Days)	Estimated Cost
Feedback on design report by organizer	by organizer	by organizer	01-Mar-2024	15-Mar-2024	15	Nil
Hardware Selection & Procurement	7	1,3,6,9,	16-Mar-2024	07-Apr-2024	22	₹ 2,10,000.00
Mechanical Fabrication & Assembly	1,3,9	4,5	07-Apr-2024	27-Apr-2024	20	₹ 60,000.00
Electronics Hardware Integration & Testing	2,8	6,7,10	07-Apr-2024	27-Apr-2024	20	Nil
Software Development & Testing	2,6	7,8,10	21-Mar-2024	27-Apr-2024	38	Nil
System Integration & Testing	2,6	4,5,7,8,10	27-Apr-2024	06-May-2024	10	Nil
Field Testing & Validation	6,7	1,3,9,10	07-May-2024	14-May-2024	7	Nil
Prototype Demonstration	All Team	All Team	15-May-2024	19-May-2024	5	Nil

Novelty in the Overall Proposal

This project demonstrates several innovative features and thoughtful engineering choices relevant to space rover design:

- **Onboard CNN for crater detection:** Efficiently identifies craters using a lightweight CNN model on the rover, minimizing computing needs.
- **Adaptive path planning:** Navigate challenging terrain (obstacles, craters, slopes) through real-time terrain mapping and obstacle avoidance.
- **Advanced Rocker Bogie:** Ensures stability and maneuverability on loose sand and inclines with a six-wheel Rocker Bogie differential mechanism.
- **5-DOF robotic arm:** Expands mission capabilities with versatile sample manipulation and placement.
- **Jetson Nano onboard computer**: Powers onboard data processing, path planning, and actuator control for independent operation. Sensor data analysis, path planning algorithms, and actuator control are crucial for autonomous operations.
- **Independent battery systems:** Optimizes performance and enhances mission criticality and redundancy by isolating potential subsystem failures.

This rover system combines innovative technologies and robust design for a successful mission on challenging terrain.